



Box Dimension of Fractals

S.N. Bhatt Presentation

Adrija Chatterjee

Project Advisor: Dr. Siva Athreya

International Centre for Theoretical Sciences

22 June 2025

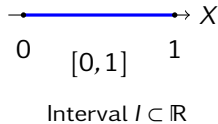
Table of Contents

1. Introduction
2. Examples
3. Box Dimension
4. Evaluating Box Dimension
5. Introduction to Some Fractals
6. A Summary of Results
7. Conclusion

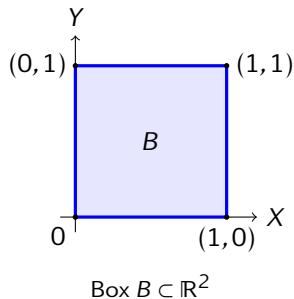
Introduction

- The dimension of a set is formally thought of as the minimum number of coordinates needed to specify any point in it.
- An interesting class of sets are fractals where the classical notion of dimension does not seem to capture their geometric properties.
- We introduce other notions of dimension, namely: topological, box, similarity and Hausdorff dimensions.
- We define fractals precisely using the above dimensions and compute them for various sets including fractals in the S.N. Bhatt project.

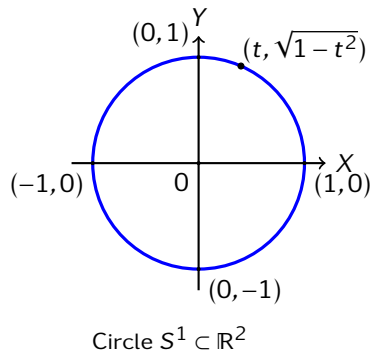
Some Standard Shapes and Their Properties



Perimeter: 1
Area: 0
Dimension: 1



Perimeter: 4
Area: 1
Dimension: 2



Perimeter: 2π
Area: 0
Dimension: 1

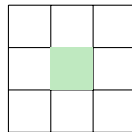
Sierpiński Carpet S_4 - designed by Wacław Sierpiński

We start with a box of side length 1 and call it U_0 .



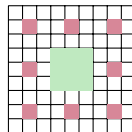
U_0

We then divide U_0 into 9 closed sub-boxes and remove the open central sub-box. This leaves us with 8 sub-boxes of length $\frac{1}{3}$ constituting U_1 .



U_1

Now, we repeat the above process for the 8 boxes of U_1 , i.e., we divide each box of U_1 further into 9 sub-boxes and remove the central open boxes. This leaves us with 8^2 sub-boxes of length $\frac{1}{9}$ constituting U_2 .



U_2

Sierpiński Carpet S_4

This process yields a sequence of sets:

$$U_0 \supset U_1 \supset U_2 \supset \dots$$

The Sierpiński Carpet is the limit of this sequence:

$$S_4 = \bigcap_{n=0}^{\infty} U_n.$$

Properties of S_4 :

- S_4 has infinite perimeter.
- S_4 has zero area.
- S_4 is connected.

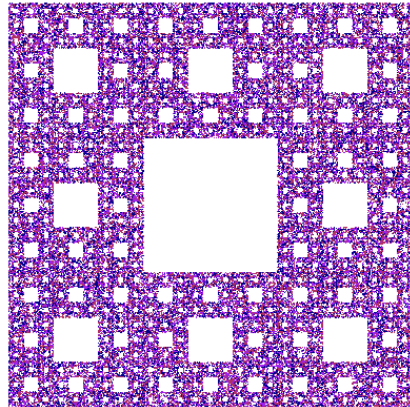


Figure 1 – Sierpiński Carpet S_4 generated in R.

Box Dimension

The Minkowski-Bouligand Dimension or Box Dimension

Let $X \subset \mathbb{R}^n$. Let $\epsilon > 0$ be given. Then $N(\epsilon)$ is the minimum number of boxes of size ϵ required to cover X . Then the **box dimension** of X ,

$$\dim_B(X) := \lim_{\epsilon \rightarrow 0} -\frac{\log N(\epsilon)}{\log \epsilon},$$

whenever the limit exists.

It is easily observed that the box-dimension of a set is a non-negative real number and could be infinity.

Box Counting on $[0, 1]^2$

To estimate the box-counting dimension of $[0, 1]^2$, we cover it using boxes of side length ϵ and count the minimum number of boxes required, $N(\epsilon)$.

ϵ	$N(\epsilon)$	$\frac{\log N(\epsilon)}{\log \frac{1}{\epsilon}}$	Covering
1	1	–	
$\frac{1}{2}$	4	2	
$\frac{1}{4}$	16	2	
$\frac{1}{2^n}$	2^{2n}	2	–

Thus for all $n \in \mathbb{N}$, we get

$$\frac{\log N\left(\frac{1}{2^n}\right)}{\log 2^n} = \frac{2n \log 2}{n \log 2} = 2.$$

We can generalize this to any $0 < \epsilon < 1$ and show:

$$N(\epsilon) = \left\lceil \frac{1}{\epsilon^2} \right\rceil \Rightarrow$$

$$\lim_{\epsilon \rightarrow 0} \frac{\log N(\epsilon)}{\log \frac{1}{\epsilon}} = \lim_{\epsilon \rightarrow 0} \frac{\log \left\lceil \frac{1}{\epsilon^2} \right\rceil}{\log \frac{1}{\epsilon}} = 2.$$

This proves that the box-counting dimension is 2.

Box Counting on the Sierpiński Carpet S_4

Let $\epsilon > 0$ be given and $N_n(\epsilon)$ be the minimum number of such boxes required to cover U_n .

ϵ	$N_n(\epsilon)$	$\frac{\log N_n(\epsilon)}{\log \frac{1}{\epsilon}}$	Covering
1	1	—	
$\frac{1}{3}$	8	$\frac{\log 8}{\log 3}$	
$\frac{1}{9}$	64	$\frac{\log 8}{\log 3}$	
$\frac{1}{3^n}$	8^n	$\frac{\log 8}{\log 3}$	—

Thus for all $n \in \mathbb{N}$, we get

$$\frac{\log N_n\left(\frac{1}{3^n}\right)}{\log 3^n} = \frac{n \log 8}{n \log 3} = \log_3 8.$$

We can generalize this to any $0 < \epsilon < 3^{-n}$ and show:

$$N_n(\epsilon) = \left\lceil \frac{\left(\frac{8}{9}\right)^n}{\epsilon^2} \right\rceil \Rightarrow$$

$$\lim_{\epsilon \rightarrow 0} \frac{\log(\lim_{n \rightarrow \infty} N_n(\epsilon))}{\log \frac{1}{\epsilon}} = \log_3 8.$$

This proves that the box-counting dimension is $\log_3(8)$.

The $\frac{1}{3}$ -Cantor Set C - designed by George Cantor

Start with the interval $[0, 1]$ and call it C_0 .



Remove the interval $(\frac{1}{3}, \frac{2}{3})$ from C_0 to get C_1 .



Remove the intervals $(\frac{1}{9}, \frac{2}{9})$ and $(\frac{7}{9}, \frac{8}{9})$ from C_1 to get C_2 .



The $\frac{1}{3}$ -Cantor Set C

Continuing this process gives us a sequence of sets:

$$C_0 \supset C_1 \supset C_2 \supset \dots$$

The Cantor set is the limit of this sequence, i.e.,

$$C = \bigcap_{n=0}^{\infty} C_n.$$

Properties of C :

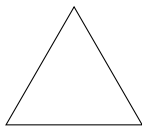
- C is uncountable.
- C has zero length.
- C is totally disconnected.
- $\dim_B(C) = \log_3(2)$.

Question - What is the dimension of C ?

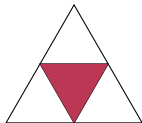


Figure 2 – The First Six Iterations of Cantor Set C .

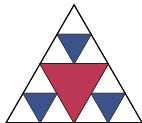
Sierpiński Gasket S_3 - designed by Wacław Sierpiński


 T_0

We start with a solid equilateral triangle of side length 1 and call it T_0 .


 T_1

We divide T_0 into four equilateral triangles and remove the central open invertex triangle to get T_1 .


 T_2

We repeat the above process for T_1 to obtain 9 equilateral triangles constituting T_2 .

Sierpiński Gasket S_3

Continuing this process gives us a sequence of sets:

$$T_0 \supset T_1 \supset T_2 \supset \dots$$

The Sierpiński Gasket is the limit of this sequence:

$$S_3 = \bigcap_{n=0}^{\infty} T_n.$$

Properties of S_3 :

- S_3 has infinite perimeter.
- S_3 has zero area.
- S_3 is connected.
- $\dim_B(S_3) = \log_2(3)$.

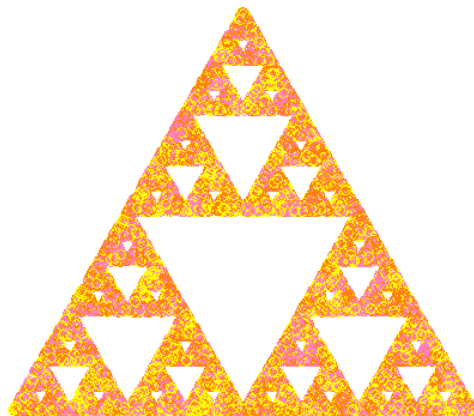


Figure 3 – Sierpiński Triangle S_3 generated in R.

Koch Curve K' - Designed by Helge von Koch

K'_0 is a line segment of length 1.

We divide the segment into three equal parts and replace the middle segment with the remaining two sides of an equilateral triangle pointing outward. K'_1 is the union of the 4 line segments, each of length $\frac{1}{3}$.

The same construction is applied to each of the 4 line segments of K'_1 to get K'_2 . K'_2 is then the union of 16 line segments each of length $\frac{1}{9}$.

This gives us a sequence of sets $\{K'_n\}_{n \in \mathbb{N}}$ and the Koch Curve K' is defined as

$$K' = \lim_{n \rightarrow \infty} K'_n.$$

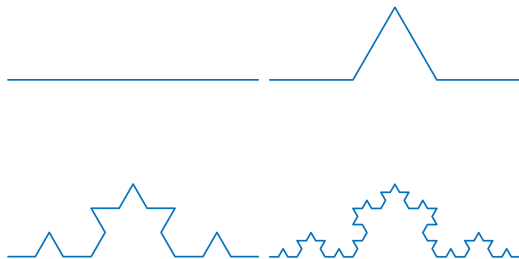


Figure 4 – First Three Iterations of the Koch Curve.

Koch Snowflake K - Designed by Helge von Koch

The **Koch Snowflake K** is created by applying the Koch Curve transformation to each side of an equilateral triangle K_0 .

Properties of K :

- K has infinite perimeter.
- K has zero area.
- K is path-connected.
- $\dim_B(K) = \log_3(4)$.

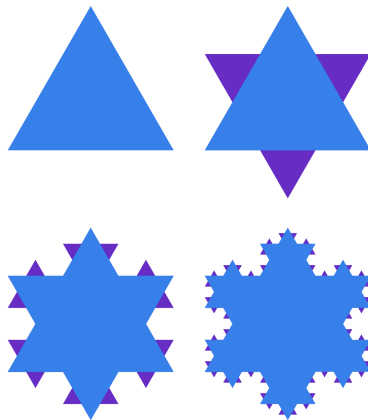


Figure 5 – The First Three Iterations of the Koch Snowflake.

A Summary of Results

This presentation only dealt with the box dimension of sets. However, other notions of dimension, namely - topological, similarity and Hausdorff dimensions have been discussed precisely in the report. Following is a summary of the sets dealt with in the project report.

Set	Perimeter	Area	Topological	Box	Similarity	Hausdorff
Interval I	1	0	1	1	1	1
Box B	4	1	2	2	2	2
Circle S^1	2π	0	1	1	–	1
α -Cantor Set C^α	0	0	0	$\log_{\frac{2}{1-\alpha}}(2)$	$\log_{\frac{2}{1-\alpha}}(2)$	$\log_{\frac{2}{1-\alpha}}(2)$
Sierpiński gasket S_3	∞	0	1	$\log_2(3)$	$\log_2(3)$	$\log_2(3)$
Sierpiński carpet S_4	∞	0	1	$\log_3(8)$	$\log_3(8)$	$\log_3(8)$
Koch snowflake K	∞	0	1	$\log_3(4)$	$\log_3(4)$	$\log_3(4)$

Table 1 – A Summary of Results.

Conclusion

We are now well-equipped to define a fractal.

We may define a “fractal” as a set whose box dimension is strictly greater than its “topological” dimension.

Notice that by this definition the Cantor Set, the Sierpiński Gasket, the Sierpiński Carpet and the Koch Snowflake are fractals.

Most fractals are characterized by non-integral box or Hausdorff dimensions.

THANK YOU!